ment in cybersecurity are needed in both the public and private sectors. Organisations need to keep up with the pace of emerging threats. There is a global shortage of people in cyber skills. Cybersecurity company Endgame created an open-source dataset to assist researchers and practitioners in training and testing of new algorithms a few years ago.

It is necessary to make employees aware of the risks too so that they do not aid cyberattacks. This is quite evident from the recent hacking of prominent Twitter accounts. As per the reports, the accused had convinced the social networking company's employees that he worked at their IT department and was a co-worker. According to a 2018 Gartner survey, nearly 88 per cent of 3,000 chief information officers (CIOs) said that cybersecurity was a top focus.

Several cybersecurity firms such as Veridu, Avaleris, WISeKey, Signifyd, Socure, and ThreatMetrix have been working on designing complete end-to-end protection solutions using the latest technologies like AI. Governments worldwide need to develop strict regulations to ensure that companies follow proper measures and laws that penalise perpetrators for their involvement in cyberattacks.

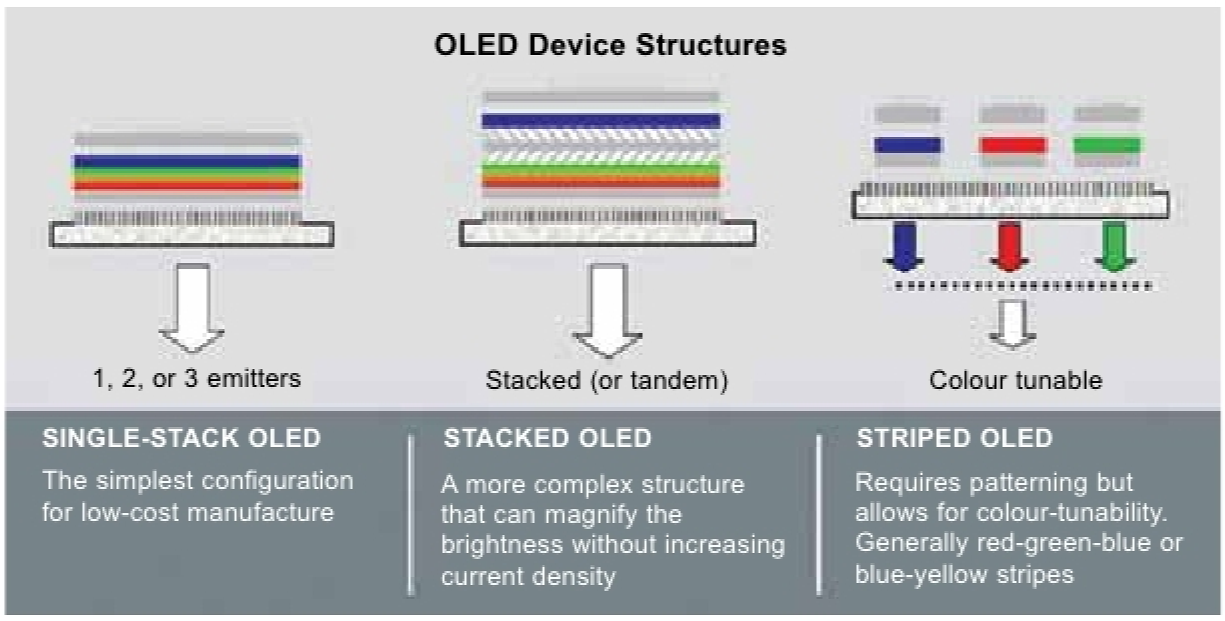
According to a report by Grand View Research Inc., the global AI market is estimated to reach 733.7 billion dollars in 2027.

—Ayushee Sharma

2 ORGANIC SEMICONDUCTORS AND LEDs

Electronics waste can result in serious environmental impact and even contribute to global warming. This is because the materials used to make electronics products are often toxic. This has led to the growth of several organic materials that have semiconductor properties for use in electronics items

Disposal of electronics waste is a huge problem. When not properly dealt with, electronics waste can result in serious environmental issues and even contribute to global warming. This is because the materials used to make electronics products are often toxic, corrosive, flammable, reactive, infectious, or even radioactive. This has led to the growth of organic materials that have semiconductor properties for use in electronics items. Unlike their inorganic counterparts, these are biodegradable, biocompatible, non-toxic and pose no threats to the environment.



(Credit: www.energy.gov)

Researches of the past few years have shown that organic semiconductors exhibit promising optical, electrical, and optoelectronic characteristics. Through the addition of different functional groups, these can be made more conducting, photoluminescent, and achieve better efficiency. These non-metallic structures can hence be easily used to manufacture economical, flexible, and green electronics for robust devices of the future, which was not possible with inorganic materials.

The Australian National University (ANU) announced the invention of a part-organic semiconductor by a team of engineers two years ago. This innovation was capable of converting electricity into light very efficiently and was seen as a way of making high-performance electronic devices such as bendable mobile phones. Bending semiconductors made of organic materials can almost double the speed of electricity flowing through them, as proved in a Rutgers University-led study.

Key players in the organic semiconductor market include LG (South Korea), BASF SE (Germany), Bayer AG (Germany), Merck & Co. (US), Samsung (South Korea), Sony Corporation (Japan), and Sumitomo Corporation (Japan). According to a report from Market Research Future, the global organic semiconductor market is expected to grow at a compound annual growth rate (CAGR) of 22.4 per cent, reaching 179.4 billion dollars by 2024.

Organic semiconductors are made up of polymers or ϖ-bonded molecules that are important active materials. These have found numerous useful applications, including organic light-emitting devices (OLEDs), organic lasers, organic thin-film transistors (OTFTs), organic solar cells, bio/chemical sensors, and much more. Among these, OLEDs have seen a huge demand worldwide in recent times as an alternative to existing lighting technologies.

Unlike conventional diodes and LEDs that use layers of n-type and p-type semiconductors, OLEDs employ organic molecules to emit electrons and holes. These consist of semiconducting organic layers stacked between two electrodes. In a basic design, there